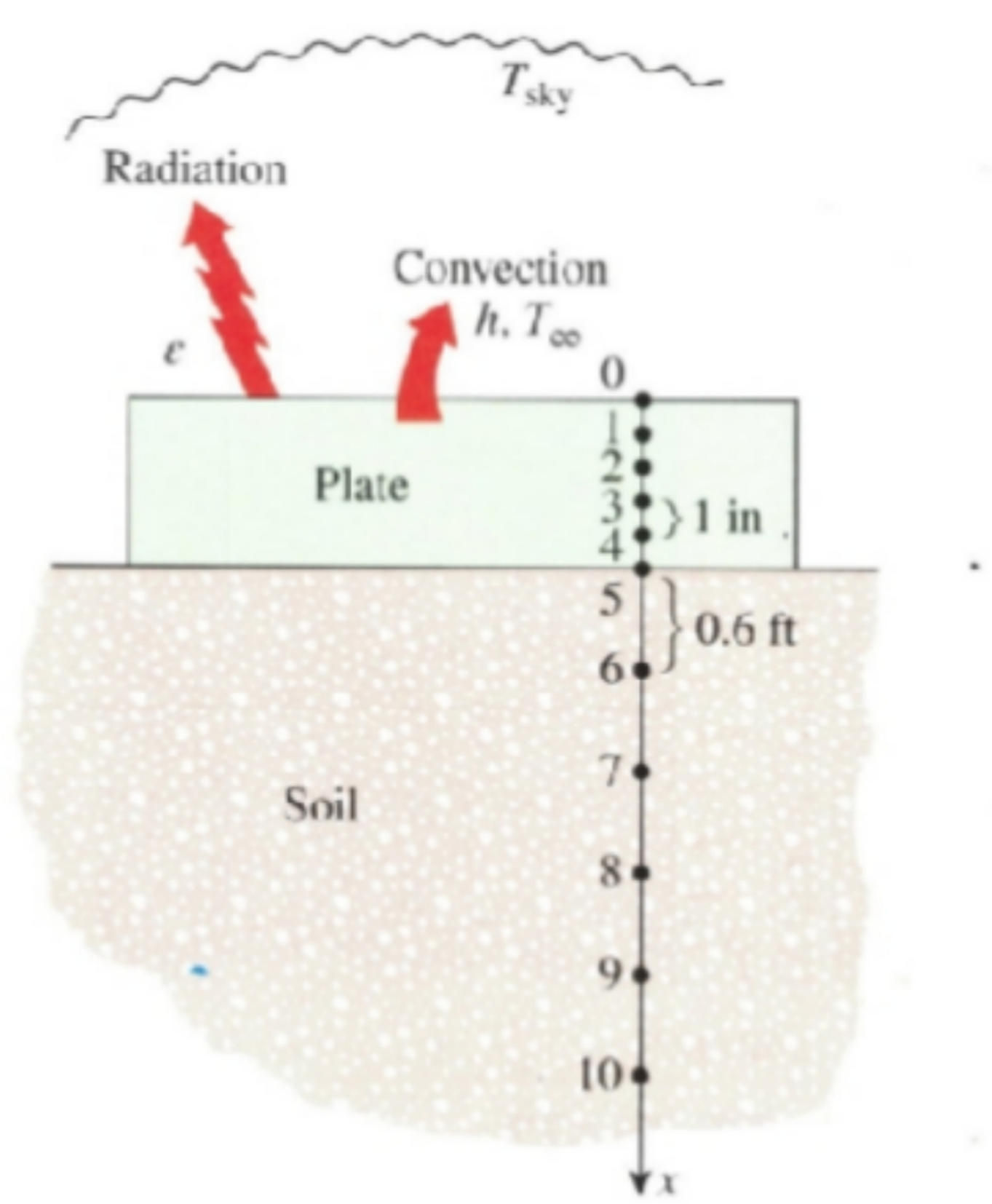
the surrounding air at $T_\infty = 15°C$ with a heat transfer coefficient of $h = 18$ W/m²·K. Assuming steady one-dimensional heat transfer and taking the nodal spacing to be 10 cm, (*a*) obtain the finite difference formulation for all nodes, (*b*) determine the nodal temperatures by solving those equations, and (*c*) evaluate the rate of heat transfer through the wall.

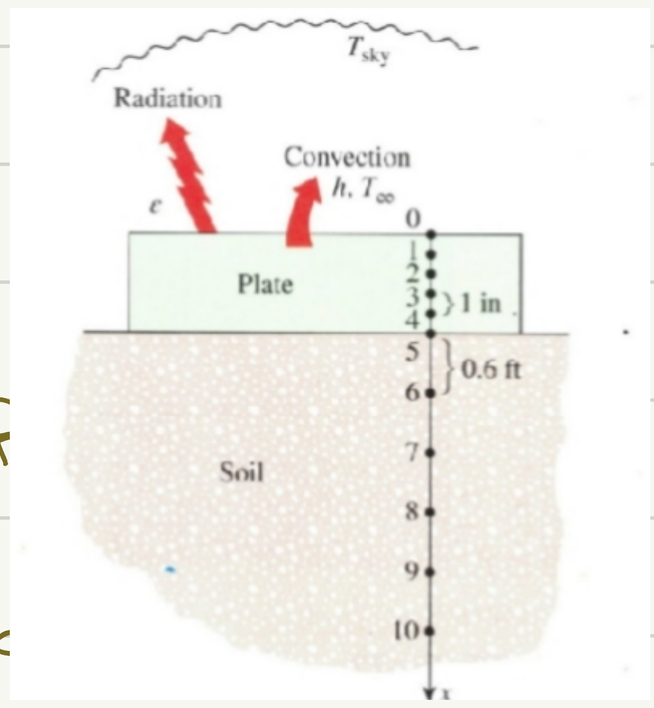
5–30E A large steel plate having a thickness of $L = 5$ in, thermal conductivity of $k = 7.2$ Btu/h·ft·°F, and an emissivity of $\varepsilon = 0.6$ is lying on the ground. The exposed surface of the plate exchanges heat by convection with the ambient air at $T_\infty = 80°F$ with an average heat transfer coefficient of $h = 3.5$ Btu/h·ft²·°F as well as by radiation with the open sky at an equivalent sky temperature of $T_{sky} = 510$ R. The ground temperature below a certain depth (say, 3 ft) is not affected by the weather conditions outside and remains fairly constant at 50°F at that location. The thermal conductivity of the soil can be taken to be $k_{soil} = 0.49$ Btu/h·ft·°F, and the steel plate can be assumed to be in perfect contact with the ground. Assuming steady one-dimensional heat transfer and taking the nodal spacings to be 1 in in the plate and 0.6 ft in the ground, (*a*) obtain the finite difference formulation for all 11 nodes shown in Fig. P5–30E and (*b*) determine the top and bottom surface temperatures of the plate by solving those equations.

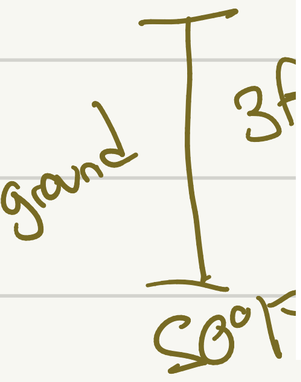


$L = 5in$ $\quad k = 7.2 \frac{BTu}{h \cdot ft \cdot F^\circ}$ $\quad \varepsilon = 0.6$ $\quad T_\infty = 80°C$

$h = 3.5 \frac{Btu}{h \cdot ft^2 \cdot F^\circ}$ $\quad T_{sky} = 510\,R$ $\quad \sigma = 0.1714 \times 10^{-8} \frac{Btu}{h \cdot ft^2 \cdot R^4}$

$k_{soil} = 0.49 \frac{BTu}{h \cdot ft \cdot F^\circ}$





a) $L = \frac{5in}{12in} \left| \frac{1ft}{} \right. = 0.417ft$ $\quad$ b) $\frac{1in}{12in} \left| \frac{1ft}{} \right. = 0.0833\,ft$

Calculando nodo cero:

- Conv
- Rad
- Cond

Ya que no sabemos la dirección de los flujos utilizamos la ecuación valida independiente de la dirección del flujo

$$q_{conv} + q_{rad} + q_{cond} = 0$$

$$h(T_\infty - T_0) + \varepsilon\sigma(T_{sky}^4 - T_{0,R}^4) + \frac{k_s}{\Delta x_s}(T_1 - T_0) = 0$$

$$q_{conv} + q_{rad} + q_{cond} = 0$$

$$h(T_\infty - T_0) + \varepsilon\sigma(T_{sky}^4 - T_{0,R}^4) + \frac{k_s}{\Delta x_s}(T_1 - T_0) = 0$$

$$3.5(80 - T_0) + 0.6(0.1714 \times 10^{-8})(510^4 - (T_0 + 459.67)^4) + 86.4(T_1 - T_0) = 0$$

sustituyendo las temperaturas resultantes de Cloud:

$$T_0 = 74.763°F$$
$$T_1 = 74.717°F$$

$$q_{conv} = 3.5(80 - 74.763)$$

$$q_{conv} = 18.329 \ \frac{Btu}{h \cdot ft^2}$$

$$q_{rad} = 0.6(0.1714 \times 10^{-8})(510^4 - (74.763 + 459.67)^4)$$

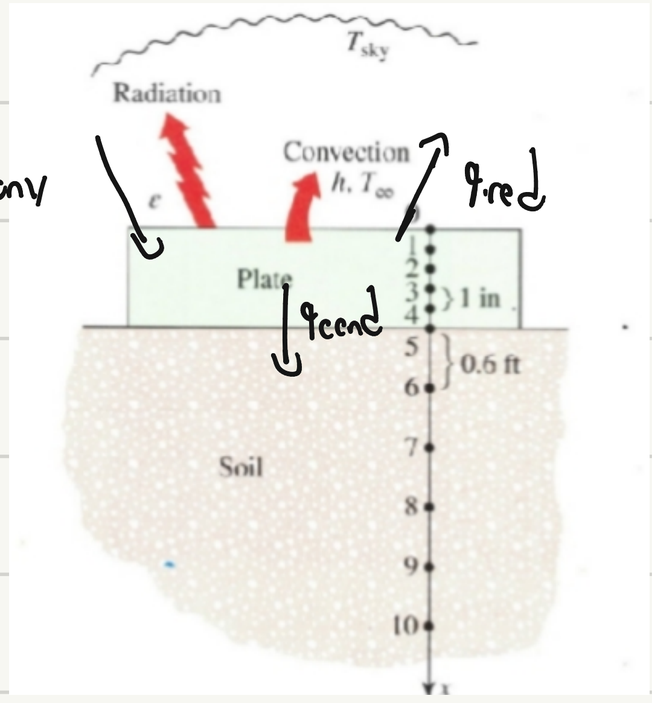
$$q_{rad} = -14.322 \ \frac{Btu}{h \cdot ft^2}$$

$$q_{cond} = 86.4(74.717 - 74.763)$$

$$q_{cond} = -3.974 \ \frac{Btu}{h \cdot ft^2}$$

$$18.329 - 14.322 - 3.974 = 0.033$$

aproximadamente 0, posiblemente por el redondeo de los cifres decimales no se alcanza cero ∴ la eaccion de Claude es correcta



q_{conv} q_{rad} q_{cond}

Nodos 1-4, dentro de la plca ∴ Conducción

- T_{i-1}
- T_i
- T_{i+1}

$\Delta x = 1 in$

$\Delta x = 1 in$

Para un nodo interior localmente:

$$\frac{K_s}{\Delta x_s}(T_{i-1} - T_i) + \frac{K_s}{\Delta x_s}(T_{i+1} - T_i) = 0$$

calor que entre al nodo i

$T_{i-1} > T_i$

calor que sale del nodo i

$T_i > T_{i+1}$

$$\frac{K_s}{\Delta x_s}(T_{i-1} - 2T_i + T_{i+1}) = 0$$

$$T_{i-1} - 2T_i + T_{i+1} = 0$$

$$T_i = \frac{T_{i-1} + T_{i+1}}{2}$$

Verificando con la T de cada nodo que nos dio Cloude:

$T_0 = 74.763$

$T_1 = 74.717$

$T_2 = 74.671$

$T_3 = 74.624$

$T_4 = 74.578$

$T_5 = 74.531$

$T_0 - 2T_1 + T_2 = 0$

Nodo 1:

$74.763 - 2(74.717) + 74.671$

$0 = 0$ ✓

Nodo 2:

$74.717 - 2(74.671) + 74.624$

$-0.001 \approx 0$ ✓

Nodo 3

$$74.671 - 2(74.624) + 74.578$$

$$0.001 \approx 0 \checkmark$$

Nodo 4

$$74.624 - 2(74.578) + 74.531$$

$$-0.001 \approx 0 \checkmark$$

∴ Perfil de temperatura dentro del acero lineal

Nodo 5

acero por encima $K_s = 7.2$

Suelo por debajo $K_g = 0.49$

Al no haber generacion de calor

$$q_{steel} = q_{soil}$$

$$q_{steel} = \frac{K_s}{\Delta x_s}(T_4 - T_5)$$

$$q_{soil} = \frac{K_g}{\Delta x_g}(T_5 - T_6)$$

$$\frac{K_s}{\Delta x_s}(T_4 - T_5) + \frac{K_g}{\Delta x_g}(T_5 - T_6) = 0$$

$$86.4(T_4 - T_5) + 0.81667(T_6 - T_5) = 0$$

$$86.4T_4 - 86.4T_5 + 0.81667T_6 - 0.81667T_5 = 0$$

$$86.4\,T_4 - 87.2167\,T_5 + 0.8167\,T_6 = 0$$

usando los T de Claude:

$$q_{steel} = 86.4\,(74.578 - 74.531)$$

$$q_{steel} = 4.0608 \quad \frac{Btu}{h \cdot ft^2}$$

$$q_{soil} = 0.8167\,(74.531 - 69.625)$$

$$q_{soil} = 4.0066 \quad \frac{Btu}{h \cdot ft^2}$$

$$q_{soil} \approx q_{steel}$$

Node G-10

$$\frac{K_g}{\Delta X_g}(T_{i-1} - T_i) - \frac{K_g}{\Delta X_g}(T_i - T_{i+1}) = 0$$

$$\frac{0.49}{0.6}(T_{i-1} - T_i) - \frac{0.49}{0.6}(T_i - T_{i+1}) = 0$$

$$0.816667\,(T_{i-1} - 2T_i + T_{i+1}) = 0$$

$$T_{i-1} - 2T_i + T_{i+1} = 0$$

T segun Claude:

$T_5 = 74.531$

$T_6 = 64.625$

$T_7 = 64.719$

$T_8 = 59.813$

$T_9 = 54.906$

$T_{10} = 50.000$

Node 6:

$74.531 - 2(64.625) + 64.719$

$0 = 0$ ✓

Node 7:

$64.625 - 2(64.719) + 59.813$

$0 = 0$

Node 8:

$64.719 - 2(59.813) + 54.906$

$-0.001 \approx 0$ ✓

Node 9:

$59.813 - 2(54.906) + 50$

$0.001 \approx 0$ ✓

Node 10

$T_{10} - T = 0$

$50.000 - 50 = 0$ ✓

Flujo de calor en el suelo:

$$q_{soil} = \frac{0.49}{3}(74.531 - 50)$$

$$q_{soil} = 4.0067 \ \frac{Btu}{h \cdot ft^2}$$

"$q_{cloud} = 4.0067 \frac{Btu}{h \cdot ft^2}$"

Ejercicio Modificado:

$$q_{conv} + q_{rad} + q_{cond} = 0$$

$$h(T_\infty - T_0) + \varepsilon\sigma(T_{surr}^4 - T_{0,R}^4) + \frac{k_s}{\Delta x_s}(T_1 - T_0) = 0$$

$$3.5(80 - T_0) + 0.6(0.1714 \times 10^{-8})(510^4 - (T_0 + 459.67)^4) + 86.4(T_1 - T_0) = 0$$

sustituyendo las temperaturas resultantes de Cloud:

$T_0 = 73.023°F$
$T_1 = 72.892°F$

$$q_{conv} = 24.495 \ \frac{Btu}{h \cdot ft^2}$$

$$q_{rad} = -13.232 \ \frac{Btu}{h \cdot ft^2}$$

$$q_{cond} = -11.232 \ \frac{Btu}{h \cdot ft^2}$$

$$24.495 - 13.232 - 11.232 = -0.0695 \ \frac{Btu}{hft^2} \approx 0 \ \checkmark$$

Nodes 1-4

$$T_{i-1} - 2T_i + T_{i+1} = 0$$

$T_0 = 73.023°F$

$T_1 = 72.893°F$

$T_2 = 72.764°F$

$T_3 = 72.634°F$

$T_4 = 72.505°F$

$T_5 = 72.375°F$

Nodo 1

$$73.023 - 2(72.893) + 72.764$$

$$0.001 \approx 0$$

Nodo 2

$$72.893 - 2(72.764) + 72.634$$

$$-0.001 \approx 0$$

Nodo 3

$$72.764 - 2(72.634) + 72.505$$

$$0.001 \approx 0$$

Nodo 4

$$72.634 - 2(72.505) + 72.375$$

$$-0.001 \approx 0$$

Nodo 5

$$q_{in} = q_{out}$$

$$\frac{K_s}{\Delta x}(T_u - T_s) = \frac{K_s}{\Delta x_g}(T_s - T_c)$$

$$86.4(T_u - T_s) + 2.5(T_c - T_s) = 0$$

$$86.4T_u - 88.9T_s + 2.5T_c = 0$$

Con los T de Claude.

$$T_u = 72.505°F$$
$$T_s = 72.375°F$$
$$T_c = 67.900°F$$

$$11.232 \frac{Btu}{h \cdot ft^2} - 11.1875 \frac{Btu}{h \cdot ft^2} = 0.0445 \approx 0$$

Nodo 6 - ϕ

$$T_{i-1} - 2T_i + T_{i+1} = 0$$

Nodo 6

$$72.375 - 2(67.900) + 63.425$$
$$0 = 0$$

Nodo 7

$$67.9 - 2(63.425) + 58.950$$
$$0 = 0$$

Nodo 8

$$63.425 - 2(58.950) + 54.475$$

$$0 = 0$$

Node 9

$$58.950 - 2(54.475) + 50$$

$$0 = 0$$

Nodo 10:

$$T_{10} - 50 = 0$$

$$50 - 50 = 0$$

Verificando flujo de calor en el web

$$q_{soil} = \frac{K}{x}(T_s - T_{10})$$

$$q_{soil} = \frac{1.50}{3}(72.375 - 50)$$

$$q_{soil} = 11.1875 \ \frac{Btu}{h \cdot ft^2}$$

% aumento:

$$\frac{q_{modificado} - q_{original}}{q_{original}}$$

$$\frac{|11.1875 - 4.0068|}{4.0068} \times 100$$

% cumento = 179.2%

Los resultados de Claude para el ejercicio modificado son consistentes con las ecuaciones de diferencias finitas y la ley de Fourier, considerando el redondeo de las temperaturas.